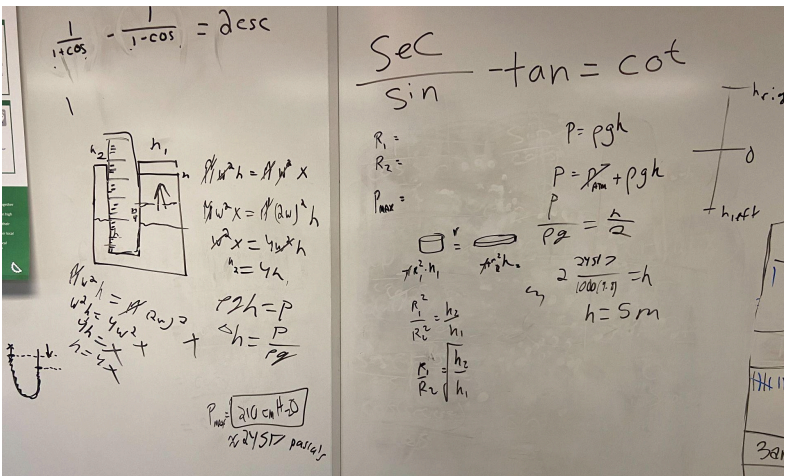
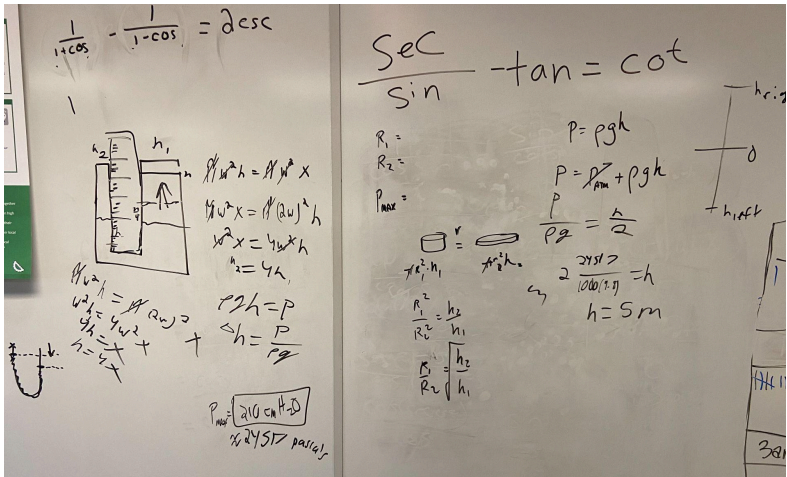
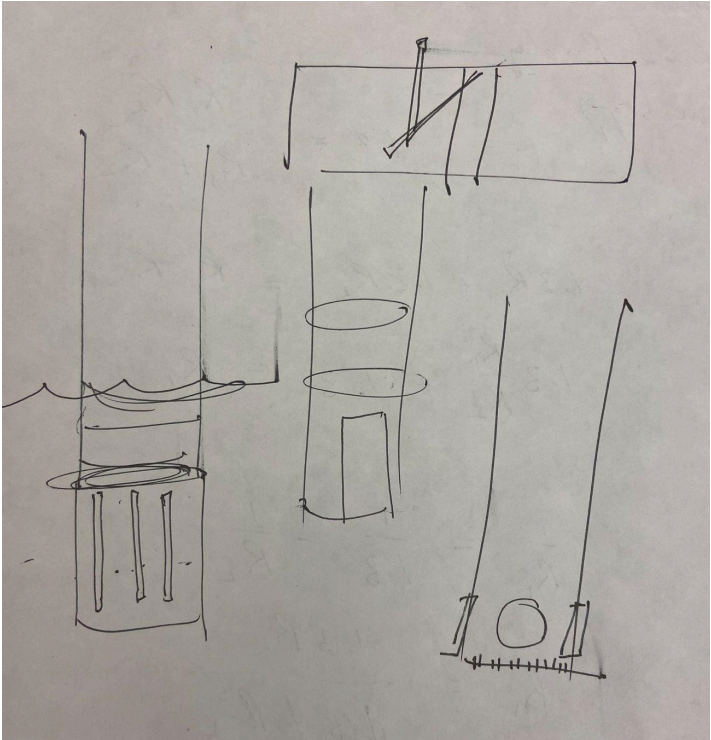


Element E

Jake Steinberg, Astrid Santiago, Caleb Bridgeman, Anne-Marie Georgi, Jack Hanson

1 (High Priority)	Safe	Projected height equation: This ensures the user does not breathe in the water during use.  Our device uses a One-way filter, often found in devices like a CPR mask. This mouthpiece ensures that the user is not exposed to any contaminants of other users without having to take apart the device after each use.
2 (High Priority)	User-friendly	We made sure that the device would be user-friendly, because the experts will not always be there in use. The cylindrical figure makes it easy to handle. We also used perfect measurements for the 3d printers, creating an easy ‘Snap on, snap off’ tactic.

		
3 (High Priority)	Precise	Projected height equation?   The picture above shows our projected height equation. Unfortunately, the equation we came down to was unsolvable so therefore we were unable to build the theoretical solution.
4 (High Priority)	Repeatability	Manual recording Continuous spiral setting calculation

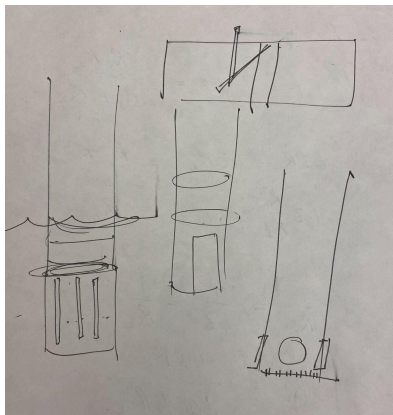
		
5 (High Priority)	Inexpensive	5 for tube 2 for mouthpiece Red 3d print: 20.8g = \$1.62 White 3d print: 33.4 grams

E 2. STEM Applications:

When the person using this device inhales through the mouthpiece, the inhaling acts like a vacuum, creating negative pressure. This results in the water rise in the clear PVC pipe. The stronger someone's inhale is, the higher the water will rise.

E 3.

We brought the initial ideas to our physics teacher, Mr. Scheidly, who took our initial ideas and designs and helped us refine them into our final design. He helped explain the idea of limiting the flow of water with the





Crescendo

We brought the prototype to my physics teacher, Mr. Flanagan, and we informed him what Dr. Brossman wanted from us. We informed him that our device does, in fact, work. However, we wanted an accurate measurement for the inhale. We first went with the “crescendo” idea. This had the bottom of the PVC pipe cut like the image to the right, which changed the resistance of the inhale. When the opening in the PVC pipe is wider, it is easier to get the water up the tube. When the clear PVC pipe is rotated, the opening of the PVC pipe decreases, making it harder to get the water to elevate to the top of the tube. Mr. Flanagan suggested using holes in the PVC pipe instead of having a gradual change. If we had a discrete number of holes, this would make calibrating easier. With limited options for resistance, it would be a lot easier to calibrate the options instead of a gradual increase/decrease.